

Introduction

Data is the foundation of Data Science and Machine Learning. The quality of data directly affects machine learning performance. This project focuses on a student academic performance dataset collected to study the relationship between study habits and academic achievement.

Data Collection Method

The dataset was collected through a questionnaire distributed to students. Information on age, gender, study hours, attendance, assignment scores, and GPA was gathered. A total of 50 student records were collected manually.

Features and Label

The dataset contains five input features: Age, Gender, Study Hours, Attendance, and Assignment Score. The target variable (label) is GPA. Features are represented as X, while GPA is represented as y.

Dataset Structure

The dataset contains 50 rows and 7 columns including Student_ID. Sample records were included to demonstrate the structure of the dataset.

Data Quality Issues

Potential issues include missing values, typographical errors, duplicate records, and class imbalance. These issues should be handled during preprocessing.

Learning Type

This is a Supervised Learning problem because a target variable (GPA) is available and can be predicted using the input features.

Use Case and Data Science Lifecycle

The dataset can be used for GPA prediction using regression models. It follows the Data Science lifecycle: problem definition, data collection, cleaning, analysis, machine learning, and deployment.

Conclusion

The dataset provides a strong foundation for learning machine learning concepts and applying predictive analytics to educational data.

References

Han et al. (2012) Data Mining: Concepts and Techniques.
Géron (2022) Hands-On Machine Learning.
Provost & Fawcett (2013) Data Science for Business.